

MITRE ATT&CK-Based; SOC Lab on Kali Linux

20 Tools & Commands for Security Monitoring, Event Correlation, Log Management & Detection Strategy
— Cyber Security Operations & SOC Management

This lab guide lists 20 open-source tools used on Kali Linux for building a MITRE ATT&CK-aligned; SOC training lab. Tools are grouped by function: network monitoring/IDS, host log collection, SIEM/correlation, threat intel & mapping, and adversary emulation (used to safely generate detectable activity for the exercises). Each entry includes install and core usage commands.

A. Network Security Monitoring / IDS-IPS

1. Suricata

Signature-based IDS/IPS/NSM engine; supports MITRE ATT&CK-tagged; rulesets (ET Open, etc.).

```
sudo apt install suricata -y
sudo suricata-update
sudo suricata -c /etc/suricata/suricata.yaml -i eth0
tail -f /var/log/suricata/eve.json | jq .
```

2. Zeek (Bro)

Protocol-aware network traffic analyzer; generates rich connection/DNS/HTTP logs used for correlation.

```
sudo apt install zeek -y
sudo /opt/zeek/bin/zeekctl deploy
zeek -i eth0 local
cat conn.log | zeek-cut id.orig_h id.resp_h proto
```

3. Snort

Classic rule-based IDS; alternative/companion to Suricata for signature detection labs.

```
sudo apt install snort -y
sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
sudo snort -T -c /etc/snort/snort.conf # test config
```

4. tcpdump

Raw packet capture for manual traffic inspection and evidence collection.

```
sudo tcpdump -i eth0 -w capture.pcap
sudo tcpdump -i eth0 port 445 -nn
tcpdump -r capture.pcap -A | less
```

5. Wireshark / tshark

GUI and CLI packet analysis for deep-dive investigation of captured traffic.

```
sudo apt install wireshark tshark -y
tshark -i eth0 -Y "http.request" -T fields -e ip.src -e http.host
tshark -r capture.pcap -q -z conv,tcp
```

B. Host-Based Logging / Endpoint Visibility

6. auditd

Linux kernel audit framework; logs syscalls, file access, and privilege changes — maps to ATT&CK; Linux techniques.

```
sudo apt install auditd audispd-plugins -y
sudo systemctl enable --now auditd
sudo auditctl -w /etc/passwd -p wa -k passwd_changes
sudo ausearch -k passwd_changes
sudo aureport --summary
```

7. Sysmon for Linux

Sysinternals Sysmon ported to Linux; detailed process/network/file event logging (mirrors Windows Sysmon).

```
git clone https://github.com/Sysinternals/SysmonForLinux.git
cd SysmonForLinux && ./build.sh
sudo ./sysmon -accepteula -i sysmon-config.xml
sudo journalctl -f -u sysmon
```

8. osquery

SQL-based endpoint instrumentation; query running processes, users, network connections like a database.

```
sudo apt install osquery -y
osqueryi "SELECT * FROM processes WHERE name='bash';"
osqueryi "SELECT * FROM listening_ports;"
sudo systemctl start osqueryd
```

9. Auditbeat

Elastic Beat that ships auditd/file-integrity data directly into the ELK/Wazuh pipeline.

```
curl -L -O https://artifacts.elastic.co/downloads/beats/auditbeat/auditbeat-8.13.0-amd64.deb
sudo dpkg -i auditbeat-8.13.0-amd64.deb
sudo auditbeat setup
sudo systemctl start auditbeat
```

10. Filebeat

Lightweight log shipper; forwards Suricata/Zeek/syslog files to Logstash/Elasticsearch/Wazuh indexer.

```
sudo apt install filebeat -y
sudo filebeat modules enable suricata zeek
sudo filebeat setup -e
sudo systemctl start filebeat
```

C. SIEM / Log Aggregation / Event Correlation

11. Wazuh (manager + indexer + dashboard)

Open-source SIEM/XDR; ships with ATT&CK-tagged; detection rules and built-in correlation engine.

```
curl -sO https://packages.wazuh.com/4.x/wazuh-install.sh
sudo bash wazuh-install.sh -a
sudo /var/ossec/bin/agent_control -l          # list agents
sudo tail -f /var/ossec/logs/alerts/alerts.json
```

12. Elastic Stack (Elasticsearch + Logstash + Kibana)

General-purpose log store/search/visualization stack for building custom correlation dashboards.

```
sudo apt install elasticsearch logstash kibana -y
sudo systemctl enable --now elasticsearch logstash kibana
curl -X GET "localhost:9200/_cat/indices?v"
```

13. Graylog

Alternative centralized log management platform with stream-based correlation rules and alerting.

```
sudo apt install docker.io docker-compose -y
git clone https://github.com/Graylog2/docker-compose.git
cd docker-compose && sudo docker-compose up -d
# Web UI: http://localhost:9000
```

14. rsyslog

Standard Linux syslog daemon; central log collection point for forwarding to SIEM.

```
sudo apt install rsyslog -y
sudo nano /etc/rsyslog.d/50-remote.conf
# add: *.* @@siem-server-ip:514
sudo systemctl restart rsyslog
```

15. Splunk Free (optional, if licensed by course)

Commercial-grade SIEM with SPL query language; many programs provide a free/dev license for labs.

```
wget -O splunk.deb 'https://download.splunk.com/products/splunk/releases/latest/linux/splunk.deb'
sudo dpkg -i splunk.deb
sudo /opt/splunk/bin/splunk start --accept-license
# Web UI: http://localhost:8000
```

D. Threat Intel Mapping / Detection Strategy

16. MITRE ATT&CK; Navigator

Interactive heatmap tool to score detection coverage per technique/tactic — core deliverable for 'detection strategy' unit.

```
sudo apt install docker.io -y
git clone https://github.com/mitre-attack/attack-navigator.git
cd attack-navigator/nav-app
docker build -t attack-navigator .
docker run -p 4200:4200 attack-navigator
# Access: http://localhost:4200
```

17. Sigma

Generic, vendor-agnostic detection rule format; write once, convert to Wazuh/Elastic/Splunk queries.

```
pip install sigma-cli --break-system-packages
git clone https://github.com/SigmaHQ/sigma.git
sigma convert -t splunk sigma/rules/windows/process_creation/*.yml
sigma convert -t elasticsearch-lucene rule.yml
```

18. MISP (Malware Info Sharing Platform)

Threat intel platform for IOC management, tagging IOCs with ATT&CK; techniques, and correlation lookups.

```
git clone https://github.com/MISP/misp-docker.git
cd misp-docker && cp template.env .env
sudo docker-compose up -d
# Web UI: https://localhost
```

E. Adversary Emulation (safe log-generation for detection exercises)

19. Atomic Red Team

Executes individual, discrete ATT&CK; techniques (by Technique ID) to safely generate detectable events.

```
git clone https://github.com/redcanaryco/atomic-red-team.git
# On Windows target (PowerShell):
IEX (IWR 'https://raw.githubusercontent.com/redcanaryco/invoke-atomicredteam/master/install-atomicredteam.ps1' -U
Invoke-AtomicTest T1059.001 # Example: PowerShell execution
```

20. CALDERA

MITRE's own automated adversary emulation platform; runs full ATT&CK; operation chains and reports technique IDs used.

```
sudo apt install python3-pip -y
git clone https://github.com/mitre/caldera.git --recursive
cd caldera && pip3 install -r requirements.txt --break-system-packages
python3 server.py --insecure
# Web UI: https://localhost:8888 (default red/blue login)
```

Suggested Lab Flow Mapped to Course Topics

Course Topic	Lab Activity	Key Tools
Log Management	Configure auditd/Sysmon; forward logs via Filebeat/Auditbeat to SIEM	auditd, Sysmon, Filebeat, Auditbeat
Security Monitoring	Deploy Suricata/Zeek on a network segment; watch live alerts	Suricata, Zeek, Snort, tcpdump
Event Correlation	Build SIEM rules combining login failures + process creation + egress to external hosts	Assimulo, Elastic Stack, Graylog
Detection Strategy	Use ATT&CK Navigator to find coverage gaps; write Sigma rules for gaps	ATT&CK Navigator, Sigma, MISP
Exercise / Validation	Run Atomic Red Team / CALDERA techniques; locate them in SIEM	Atomic Red Team, CALDERA

Note: All 20 tools listed are open-source/free and defensive or MITRE-sanctioned adversary-emulation tools appropriate for a monitoring/detection/SOC coursework lab. Run attack simulation tools (Atomic Red Team, CALDERA) only in isolated lab environments you own or are authorized to test.